From Formulation to Articulation: Considerations for AI-Assisted Research in Information Theory

Abstract

Recent advances in artificial intelligence have contributed to several information-theoretic results through genuinely different modes: a language model may suggest a proof, participate in an extended mathematical dialogue, coordinate computation, or operate as one component of a search harness. Treating all of these as "AI-discovered mathematics" obscures distinctions that matter for scientific reliability, attribution, and reproducibility. This article examines a set of recent cases through a four-stage pipeline: how the problem was formulated; how AI participated in finding a solution; how the resulting analytical and computational claims were verified; and how machine-generated outputs were translated into mathematical exposition for human understanding. Based on these cases, we recommend that research publications (1) disclose the nature and extent of AI assistance at each stage, (2) report verification of analytical reasoning separately from certification of numerical computations, and (3) place more emphasis on articulating the underlying ideas, techniques, and interpretations.

1 Introduction

Information theory is unusually fertile ground for AI-assisted mathematics. Its problems mix inequalities, probability, discrete optimization, coding constructions, and computations that can often be certified exactly. The same mixture, however, makes broad statements such as "the AI proved the theorem" particularly ambiguous. A model might have supplied the decisive lemma, generated a proof candidate later reconstructed by people, or merely completed tedious but otherwise routine derivations. As AI models become more capable, it is important to examine not only whether they contribute to research, but also how they affect the entire process from problem formulation to publication.

We view the research process in information theory, from problem formulation to articulation of the results, as a pipeline with four stages. For these stages, we ask the following questions:

- **Problem formulation:** Is the question a longstanding conjecture, a question posed by a human researcher, or a direction formulated or materially sharpened by AI?
- **AI-mediated solution:** Is the model used through a short prompt, an interactive collaboration, or an engineered harness involving search, tools, agents, and repeated evaluation?
- **Verification:** Who checks the analytical reasoning, and what kind of numerical certification is supplied?
- **Translation and articulation:** Is the final argument reconstructed and written by human experts, revised from an AI-generated draft, or left close to its machine-generated form?

Most of the works discussed below are recent preprints, and both their proofs and their descriptions of AI involvement may change. The classifications are therefore assessments of the *documented process*, not judgments about private work that may have occurred but was not publicly reported.

At present, AI is changing the solution stage most directly, with consequential effects on verification and exposition. Whether its influence will extend substantially to problem formulation remains unclear.

Historically, research publications have served several roles simultaneously: (1) preserving an archival record of scientific progress; (2) presenting sufficient evidence for claims to be evaluated and reproduced; (3) articulating novel ideas, techniques, and interpretations so that researchers can understand and build upon them; and (4) establishing priority and attributing credit. As AI-assisted research becomes more prevalent, these functions may no longer be served entirely by a single conventional paper. AI systems may increasingly evaluate and extend results using detailed records, which include code, computational certificates, interaction histories, and intermediate artifacts. These records may be too extensive or insufficiently organized for direct human consumption. Scientific publications may therefore need to evolve in two complementary directions: preserving a sufficiently complete and machine-readable research record for reproduction and further discovery, while providing a human-oriented exposition that explains the essential ideas, techniques, evidence, significance, and limitations. An important challenge will be to maintain a reliable correspondence between these two layers, so that the human-readable account accurately represents what is supported by the underlying research record. Moreover, as the generation of new results becomes increasingly automated, the traditional emphasis on establishing individual priority and assigning credit for the act of discovery may diminish, while credit for problem formulation, verification, interpretation, and exposition may become more salient.

The human-oriented layer remains important for reasons that extend beyond the immediate production of further scientific results. AI-assisted research is likely to have substantial societal impact in the near future, and society has a legitimate interest in knowing when and how AI systems contributed to consequential scientific developments, how the resulting claims were verified, and to what extent they were understood and endorsed by human experts. Although the general public cannot be expected to evaluate technical arguments directly, it relies on researchers and scientific institutions to scrutinize, interpret, and communicate them. The cases studied below reveal a wide spectrum of AI-use patterns and human-review practices, some of which support these purposes more effectively than others. Based on this analysis, we make three recommendations. First, publications should disclose the nature and extent of AI assistance at each stage of the research pipeline. Second, verification of analytical reasoning should be reported separately from certification of numerical computations. Third, reproducibility alone is insufficient: publications must provide a human-oriented articulation of the underlying ideas, techniques, interpretations, significance, and limitations, so that the results can be understood and integrated into broader scientific theory rather than merely reproduced or extended through automated means.

It should be noted that since information theory is a subject of applied mathematics, the situation information theorists face is not unique. In fact, Tao's recent article partially influences our view [1]. The difference is that, at the time, Tao looked ahead under the assumption that a capable AI system would eventually emerge, whereas in information theory research, current AI models have already produced multiple meaningful and significant results, as our case studies detail in the sequel.

2 From Research Formulation to Articulation: A Taxonomy

This section introduces a taxonomy for the four stages identified above. The purpose is to describe the documented research process, rather than to assess the importance of a result or assign credit between humans and AI. The dimensions are intentionally kept separate: for example, a highly

agentic solution process need not imply weak verification, and a human-written exposition need not imply that every step was independently checked. Mixed workflows may exhibit more than one mode; in such cases, the classification should identify the mechanism that made the material contribution at each stage. The taxonomy is defined here without reference to particular works and is applied to the case studies in Section 4.

2.1 Problem Formulation

Problem formulation includes identifying a question, delimiting its scope, and explaining why its resolution would be consequential. This stage should be classified according to the origin of the decisive formulation, not according to the age or difficulty of the problem. We distinguish three levels of AI involvement:

- **F-0**: Human-formulated. The target theorem, conjecture, construction, or counterexample problem is supplied to the AI system with sufficient specificity that the system's task is to seek a solution.
- **F-1**: AI-refined. A human supplies a domain, broad objective, or initial question, while the AI system proposes a consequential subproblem, reduction, conjecture, or reformulation that materially changes the research direction.
- **F-2**: AI-originated. Given only broad domain-level guidance, the AI system identifies a research question and provides a substantive account of its novelty and significance before the question is selected for further investigation.

The distinction between **F-1** and **F-2** depends on both the initial human framing and the process by which the final question is selected. Merely restating a human-provided problem, generating a nearby variant, or supplying motivation after the fact does not constitute an AI-originated formulation. An **F-2** classification therefore requires unusually detailed documentation of the interaction history and of the system's role in recognizing significance.

2.2 AI-mediated Solution

The solution stage concerns how AI contributes after a research question has been selected. The phrase "using an LLM" spans at least three modes, distinguished primarily by the interaction and orchestration mechanism:

- **S-0**: Direct prompting. A researcher provides the problem, possibly together with references or a partial derivation, and receives a candidate idea, construction, or proof. Brief follow-up prompts may occur, but no sustained collaborative process or external orchestration layer is documented.
- **S-1**: Interactive collaboration. The AI system and human researchers repeatedly exchange conjectures, examples, objections, computations, and revised arguments. Humans provide feedback, select promising directions, detect errors, and reprioritize subtasks as the solution develops.
- **S-2**: Harnessed search or agentic workflow. The AI system operates within a structured loop involving one or more components such as program execution, evaluators, search or population management, external tools, multiple agents, proof-state tracking, or automated iteration.

These modes also reflect the nature of the researcher's involvement. A direct prompt may follow substantial human work in formulating a precise question, while an interactive collaboration

may require researchers to supply intermediate insights, diagnose failed arguments, and guide successive attempts. A harness shifts much of the researcher's effort toward designing the search space, evaluators, and computational workflow. The difficulty and structure of the problem therefore influence which mode is useful and how much human reasoning it requires. In assessing a result, the solution mode should be considered alongside the documented human contributions and the balance between conceptual reasoning and computation; none of these alone determines intellectual credit or reliability. Reporting such mode will help fellow researchers who would follow up on the reported research results, either to reproduce, or to develop further results.

2.3 Verification

Verification concerns whether the published claim is adequately supported, independently of how the candidate solution was generated. AI assistance makes the verification process especially important to document, but it does not by itself transfer responsibility away from the authors of the published work. A further distinction is essential: checking the analytical reasoning is different from certifying a finite computation produced after an analytical reduction. We use the following scale for the analytical-reasoning component:

- **A-0**: No documented analytical verification. The public record does not say that a human expert or proof assistant checked every indispensable analytical or reasoning step.
- **A-1**: Author verification. The authors report checking or reconstructing the complete reasoning chain. Merely accepting responsibility for the paper, without documenting such a check, does not by itself establish this level.
- **A-2**: Independent expert verification. A qualified expert outside the proof-generation loop checked the submitted reasoning and its indispensable reductions.
- **A-3**: Formal verification. The analytical argument has been formalized in Lean or another sound proof assistant and checked relative to an explicitly stated trusted base.

In traditional research papers, author verification is generally presumed: the authors developed the result, present its proof, and take responsibility for its correctness. This convention corresponds broadly to **A-1**, although it does not tell readers precisely which steps the authors checked or how they checked them. When an independent expert has examined the indispensable steps of a particular proof, that contribution may support **A-2**; an acknowledgment should make the scope of the examination clear. With AI-assisted research, these two levels are retained, and should be made more explicit in the publication.

The distinction between **A-0** and **A-1** becomes especially important when AI systems generate substantial parts of an argument. **A-0** means that verification of the analytical reasoning has not been documented; it does not establish that no verification occurred. Nevertheless, we believe that scientific publication of a claimed theorem should generally require at least **A-1**: the authors should be able to verify the indispensable analytical steps and explicitly attest that they have done so. A rigorous numerical certificate cannot replace this requirement when the reduction connecting the certificate to the theorem has not been verified. Only in exceptional circumstances should a result with **A-0** status be considered acceptable, and then its unverified status should be stated prominently rather than presenting the claim as an established theorem.

Formal verification (**A-3**) offers another route to establishing correctness, including when the authors cannot independently check every step of a long proof. It establishes the formalized claim

relative to its stated definitions, assumptions, and trusted base. The correspondence between that claim and the theorem described in the paper must still be checked. Moreover, a formally verified claim does not by itself explain the ideas behind the result or support additional assertions made only in the human-readable exposition.

For results that depend on numerical optimization, root solving, exhaustive search, or a computationally verified counterexample, we use a separate numerical scale:

- **N-0**: Not used or not applicable. The proof is completely analytical, or numerical work is only exploratory and carries no logical load for the correctness of the result.
- **N-1**: Ordinary numerical evidence. Floating-point experiments, Monte Carlo results, or plots support the claim without a rigorous enclosure.
- **N-2**: Rigorous certificate. Exact rational arithmetic, interval or ball arithmetic with outward rounding, an exhaustive finite check, or a mechanically checkable witness discharges a clearly specified finite obligation.
- **N-3**: Independently replayed or formally integrated certificate. The certificate was reproduced independently, or its checker and connection to the theorem were formally verified.

For example, a computer-assisted proof may follow the workflow

$$\text{theorem claim} \xrightarrow{\text{analytical reduction}} \text{finite obligations} \xrightarrow{\text{certified computation}} \text{verified obligations.} \quad (1)$$

An **N-2** certificate establishes only the second step. If the analytical reduction remains at **A-0**, the overall theorem remains provisional. More generally, the overall assessment is controlled by the weakest indispensable link, rather than by the strongest artifact in the work.

2.4 Translation and Articulation

The final stage concerns how machine-generated ideas, derivations, computations, and artifacts are converted into an account intended for human understanding. Translation and articulation mean more than stylistic editing: reconstruction can expose missing quantifiers, hidden regularity assumptions, or a mismatch between a program and the theorem it is supposed to establish. We distinguish three modes according to the documented degree of human reconstruction:

- **T-0: Machine-led presentation.**The argument or its exposition is substantially AI-generated, with no documented substantive human reconstruction beyond direction, feedback, or final review.
- **T-1: Author-revised AI exposition.** Substantial prose or derivations originate in AI output, while the authors review, edit, reorganize, and clarify the material for publication.
- **T-2: Human-led articulation.** Human researchers construct or substantially reorganize the final exposition around the underlying ideas, techniques, interpretations, significance, and limitations. AI may have contributed to the mathematics or assisted with drafting. This classification concerns the presentation and does not, by itself, establish analytical verification.

A formal proof or machine-readable research record is complementary to this scale and should be described through the verification and reproducibility record; it does not by itself provide human-oriented articulation. Conversely, polished exposition does not establish correctness. Translation may contribute to author verification by exposing errors during reconstruction, but the **T**- and **A**-levels should remain separately reported.

3 Granularity and Attribution of AI Contributions

Two related questions arise when the contributions of humans and AI systems are intertwined: what should be treated as the unit of analysis, and whether those contributions should be quantified.

3.1 Hybrid Work with Distinct Components

The classifications in Section 2 need not apply uniformly to every part of a research article. At one extreme, human researchers may formulate a theorem and develop most of its proof, while using AI to establish one lemma or overcome a decisive obstacle. At the other extreme, a largely AI-generated argument may depend on a crucial human reduction or correction. Describing either work simply as "human-proved" or "AI-proved" would obscure its development and make it harder for future researchers to identify and build on the principal ideas.

When distinct contributions can be identified, the appropriate unit of analysis may be a substantive component, such as a reduction, lemma, construction, counterexample, proof, or finite certificate. Authors should describe the principal components developed with AI assistance, the mathematical work that preceded that assistance, and how those components were incorporated into the final result. A short lemma may warrant prominent disclosure if it resolves the principal obstacle to a theorem; its length in the finished paper does not measure its importance.

In a sustained interactive collaboration, the development of individual components may be too intertwined to reconstruct in this way. A paper-level account may then be more appropriate, provided it describes the respective roles of the researchers and the AI system and identifies any decisive contributions that can be distinguished. The aim is an informative account of how the result was obtained, not an exhaustive attribution of every proof step.

Origin and verification must be kept distinct. If AI supplied a key lemma and the authors subsequently checked both its proof and its role in the theorem, the lemma remains an AI-assisted contribution, while analytical verification of the complete argument may reach **A-1**. Human verification does not change the origin of the argument. Conversely, a human-produced proof that relies on an unchecked AI-generated lemma does not reach **A-1** for the complete theorem.

3.2 Quantification of Contributions

Should a percentage split between human and machine contributions be indicated? We believe this can be extremely difficult to quantify, and also unnecessary. Such a number lacks a stable interpretation unless the quantity being measured is specified. Time spent, amount of text, number of proof steps, originality, and importance to the result can yield very different estimates, some of which can be highly subjective measures. Contributions may also be interdependent: a human-formulated reduction may enable an AI-generated lemma, which in turn enables the remainder of a human proof.

A more informative disclosure describes the respective roles of humans and AI systems and, where identifiable, the origin, development, and verification of the principal components, as outlined in the previous section. Such an estimate should not replace the account of the research process. Authors may additionally assess which contributions were decisive, explaining the basis for that assessment. If they choose to give a numerical estimate, they should specify what it measures. The reasoning/computation estimates used later in this article describe the character of the mathematical work; they do not divide that work between humans and AI.

4 Case Studies

We now apply the taxonomy to public descriptions of recent AI-assisted results. The subsections group cases by the reported mode of AI participation. These groupings are necessarily approximate, and a single paper may contain components developed in different ways. Throughout, we distinguish reported facts from our assessment of the workflow and its documented verification.

4.1 Counterexample Discovery with AI-Assistance

Gaussian completely monotone conjecture. Gu and Sellke [2] give an explicit probability measure whose Gaussian-smoothed entropy violates the relevant complete-monotonicity property, disproving the Gaussian completely monotone conjecture posed ten years earlier [3]. It was reported that GPT-5.5 Pro found the counterexample. The published object is then analyzed by conventional argument, with SageMath/Arb computations using exact inputs and rigorous ball arithmetic. Because the key advance is the structural counterexample rather than a large search, we estimate an 80/20 reasoning/computation split. The public disclosure does not establish whether the model produced the example after a direct prompt or a longer interaction.

Fisher information along the heat flow. Zou et al. construct a smooth positive density in two dimensions that violates log-convexity of Fisher information along heat flow [4], disproving the same complete-monotonicity property in higher dimensions. Their revised abstract describes GPT-5.5 Pro-assisted exploration. The final proof uses a hexagonal perturbation on a triangular torus and an analytical transfer to a Gaussian-decaying density; the Fourier-grid calculation provides supporting evidence rather than carrying the theorem on its own. We assess the result as predominantly reasoning-driven (roughly 70/30), with author-presented analytical verification and supplementary numerics. The precise interaction with the model is not disclosed.

4.2 Interactive and Guided Proof Development

Bernoulli measures conditioned on affine subspaces. Wu resolves an Ancheta–Massey-type entropy question for Bernoulli parameters below one half [5]. The paper states that the proof was discovered by GPT-5.6 Sol under interactive guidance. It also records the translation step: the author simplified the program, developed a self-contained proof, wrote the paper, and accepted responsibility. This is an example of an AI-assisted route that the human author subsequently reconstructed. The proof is essentially analytical; we estimate a 90/10 reasoning/computation split and assess verification as **A-1**/**N-1**.

Marton's inner bound. Huang et al. showed that Marton's inner bound is not generally optimal for the two-receiver broadcast channel [6, 7]. The acknowledgments describe Chandra Nair proposing an AI-assisted counterexample search at ISIT 2026. Several model-assisted stages followed, including a Markovity counterexample, geometric elimination, ternary examples, and refinements from constrained to unconstrained statements. The reported models include GPT-5.6 Sol and Claude systems. Nair and Amin Gohari are reported to have checked technical nonnumerical aspects and assisted with organization and writing. Rational distributions and interval arithmetic support the computational obligations. On the reported scope of those checks, this is an **A-2**/**N-2** case: expert scrutiny addresses the analytical steps, while rigorous computation addresses the finite instances. We estimate a 70/30 reasoning/computation split.

Second-order asymptotics for the Gaussian MAC. In two connected preprints, Tan characterizes the second-order coding-rate region of the two-user Gaussian multiple-access channel at its corner points [8] and at the relative interior of the sum-rate face [9]. The latter work shows that, for error probabilities below one half, a suitable mixture of power splits improves on constant-power inner bounds, and establishes a matching converse.

Both manuscripts disclose substantive assistance in exploring and stress-testing proof strategies, formulating and refining intermediate lemmas, checking calculations and possible gaps, and refining exposition. The corner-point paper names OpenAI Codex (GPT-5.6 Sol); the subsequent paper names OpenAI Codex (GPT-6 Astra) and Claude (Opus 5). Tan states in each manuscript that he independently verified every mathematical statement and proof in the final version. This is unusually explicit documentation of **A-1** verification. The disclosures do not isolate a particular AI-generated lemma or give enough interaction history to quantify the model's contribution to each proof. The results are predominantly analytical; their illustrative numerical work does not appear to carry an indispensable proof obligation (**N-1**). We estimate a 90/10 reasoning/computation split, rather than a human/AI split.

The Courtade–Kumar conjecture and multibit extensions. Mahdavifar and Beirami claim a proof of the Courtade–Kumar conjecture for every dimension and binary symmetric-channel parameter, together with multibit extensions [10]. Their disclosure distinguishes two components of the same paper that should not receive a single description of AI involvement.

For the *single-bit theorem*, the longstanding question is human-originated, while the disclosure states that generative AI systems produced the proof and its detailed derivations under the authors' high-level direction and feedback. The argument combines entropy flow, boundary inequalities, Fourier analysis, and logarithmic Sobolev estimates; exact-arithmetic certificates address selected finite inequalities. We estimate an 80/20 reasoning/computation split. The authors report reviewing the complete manuscript and accepting responsibility, but do not explicitly report independently checking or reconstructing every indispensable analytical step. Under the documented-verification criterion of Section 2, the submitted derivation therefore remains at **A-0**, with **N-2** for its stated finite certificates. This classification concerns what the disclosure establishes, not whether the derivation is correct.

Two independently posted works also claim proofs of the same single-bit theorem: Chen et al. give a computer-assisted differential-equation/Bellman argument [11], while Ky and Tran give a different entropy-production and spectral argument [12]. These provide theorem-level corroboration, but do not verify each step of the Mahdavifar–Beirami derivation.

For the *multibit extension*, the authors state that they formulated the problem, selected the code constructions and computational questions, and interpreted the evidence and conjectures. They claim that shortened-Hamming quantizers beat the coordinate benchmark for output sizes at least eight, with a direct-sum extension. Structured-code searches and Monte Carlo experiments found no violation for output sizes two through seven, motivating a conjecture in that range. This component is thus a human-directed hybrid of construction, proof assistance, and computational exploration, approximately 55/45 in reasoning/computation. Its proved construction claims and its numerically motivated conjecture should be assessed separately; evidence from Monte Carlo search does not certify the conjecture.

4.3 Harnessed Search and Agentic Proof Production

Deletion-correcting codes. Weindel and Heckel adapt a FunSearch-style evolutionary loop to deletion-correcting codes [13]. An LLM repeatedly proposes scoring or construction programs;

Result	F	S	A	N	T	R/C
Gaussian monotonicity counterexample [2]	**F-0**	–	**A-1**	**N-2**	**T-2**	80/20
Fisher-information counterexample [4]	**F-0**	–	**A-1**	**N-0**	**T-2**	70/30
Bernoulli affine-subspace entropy [5]	**F-0**	**S-1**	**A-1**	**N-0**	**T-2**	90/10
Marton inner-bound counterexample [7]	**F-0**	**S-1**	**A-2**	**N-2**	**T-2**	70/30
Gaussian MAC corner points [8]	**F-0**	**S-1**	**A-1**	**N-0**	**T-2**	90/10
Gaussian MAC sum-rate face [9]	**F-0**	**S-1**	**A-1**	**N-0**	**T-2**	90/10
Courtade–Kumar, single-bit proof [10]	**F-0**	**S-1**	**A-0**	**N-2**	**T-0**	80/20
Courtade–Kumar, multibit extension [10]	**F-0**	**S-1**	**A-0**	**N-2/N-1**	**T-2**	55/45
Deletion-correcting codes [13]	**F-0**	**S-2**	**A-1**	**N-0/N-2**	**T-2**	15/85
Binary deletion-channel capacity [14]	**F-0**	**S-2**	**A-0**	**N-2**	**T-0**	55/45
Champion codes [16]	**F-0**	**S-2**	**A-1**	**N-2**	**T-2**	10/90

Table 1: Comparison of AI-assisted information-theoretic results. Short descriptions and citation numbers replace full titles; complete references appear in the bibliography. R/C denotes the estimated reasoning/computation balance, not the human/AI contribution.

a conventional evaluator ranks them and retains useful variants. For one deletion, a discovered function is converted into a theorem recovering a conjectured-optimal Varshamov–Tenengolts code. For more deletions and larger alphabets, the gains are principally improved constructions and empirical heuristics. The paper also reports that sampling more candidate functions was more useful than allocating the same resources to longer reasoning traces. The workflow is predominantly search and optimization (about 15/85 reasoning/computation), with an analytical theorem extracted for one part of its output.

Binary deletion-channel capacity. Papailiopoulos claims explicit upper and lower bounds locating binary deletion-channel capacity to within one hundredth of a bit [14]. The project used GPT-5.6 Sol, GPT-6 Astra, and Claude Fable in coding-agent environments. According to the disclosure, agents developed arguments, designed and ran numerical checks, reviewed proofs, drafted and revised the manuscript, and exchanged intermediate results; the author posed the question, steered the direction, and directed the presentation. A repository provides large proof inputs and code for exact or interval checks [15].

The finite checks are designed as certificates rather than plots, supporting **N-2**. The public disclosure, however, does not document a human expert checking every indispensable analytical reduction, and the manuscript is reported as substantially agent-generated. Its documented analytical status is therefore **A-0**. We describe the overall claim as a provisional AI-generated computer-assisted proof with certified numerical premises: the certificates cannot by themselves verify the reduction connecting those premises to channel capacity. We estimate a roughly 55/45 reasoning/computation split, with substantial uncertainty.

5 Comparison and Observations

5.1 Comparison

Table 1 summarizes the documented research pipeline. The columns record problem formulation (**F**), AI-mediated solution (**S**), analytical verification (**A**), numerical certification (**N**), and translation (**T**). A dash means that the public description does not support a specific classification. The final column gives a coarse reasoning/computation estimate. It does not measure the division of work or credit between humans and AI.

Several entries require claim-specific interpretation. For the multibit extension, **N-2** refers to finite construction checks, whereas **N-1** describes numerical evidence motivating the small-output conjecture. For deletion-correcting codes, the one-deletion theorem is analytical (**N-0**), while finite checks support other construction claims (**N-2**). The supplementary grid calculation in the Fisher-information paper carries no proof obligation and is therefore classified as **N-0**.

The **T** levels assess the final presentation, not who first found the proof or whether its analytical steps were verified. **T-2** denotes a human-led exposition of the ideas and results; Wu's reconstruction is explicitly documented, while some other assignments are assessments of the published presentation. The single-bit Courtade–Kumar proof and deletion-channel capacity claim are classified as **T-0** because substantial human reworking of their AI-generated arguments is not documented. A **T-2** classification does not, by itself, establish **A-1**.

The **A-0** entries indicate that a complete analytical check is not documented; they do not assert that no such check occurred. Independent claimed proofs of the Courtade–Kumar statement [11, 12] provide theorem-level corroboration without verifying the particular Mahdavifar–Beirami derivation. None of the listed analytical arguments is reported as fully formalized (**A-3**).

5.2 Observations Across the Pipeline

Problem formulation remains largely human-led. In the cases surveyed here, the consequential research question was formulated or selected by human researchers. Indeed, every case in Table 1 is classified as **F-0**. AI systems have contributed to finding solutions and sometimes to sharpening intermediate questions, but the available accounts do not establish an AI-originated research problem in the sense of **F-2**. This observation concerns the present sample, not a limit on what AI systems may be able to formulate in the future.

Human articulation is common, but its future form is open. Most of the surveyed works involve substantial human effort in presenting and interpreting the result. Authors have converted AI-assisted discoveries into proofs, explained constructions, and distinguished established claims from numerical evidence and conjectures. Such work can expose gaps and, more broadly, make a result intelligible to other researchers.

It is less clear whether comparable exposition will always require authors to perform this translation themselves. An author might instead direct an AI system, through a sequence of prompts, to explain the underlying ideas and organize the final presentation. We would classify a substantively guided and reviewed process of this kind as **T-1**. The resulting text might be clear, but its clarity alone would not show how well the authors understand the argument or whether the exposition faithfully represents the verified result. Those questions should be addressed separately from the quality of the prose.

Solution and verification practices are more varied. The cases exhibit direct model assistance, extended mathematical interaction, evolutionary search, and coding-agent workflows. They also differ in how analytical arguments and finite computations are checked. These choices reflect the structure of the problem, the suitability of available tools, and the researchers' methods and preferences. Consequently, an account of AI involvement is most informative when it identifies the solution mechanism and the verification performed for the resulting claims, rather than assigning a single label to the paper.

6 Conclusion

The cases surveyed here show several ways in which AI can assist information-theoretic research, rather than a single model of AI-generated mathematics. At present, human researchers still formulate or select the consequential questions in the documented cases. The solution stage is more varied, ranging from model-assisted counterexample discovery and mathematical dialogue to harnessed search and agentic proof production. A single paper may also combine human- and AI-developed components that deserve separate description.

Verification must follow the logical structure of each claim. Certification of finite computations does not establish the analytical reduction that makes them relevant, and corroboration of a theorem by another proof does not check a particular derivation. Publications should therefore state who verified the indispensable analytical steps and what the numerical certificates establish.

Most works in this sample also involve substantial human effort in articulating their results. AI may increasingly perform parts of that translation under authors' guidance, but a clear final text alone does not establish human understanding or endorsement of its claims. Scientific communication should preserve both an auditable research record and an account that enables people to understand the ideas, evidence, and limitations. Clear disclosure of AI's role throughout the pipeline will help researchers and the broader public assess what was discovered, how it was checked, and what human experts can explain and stand behind.

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